

⚡ PHYSICS CH.4 — ELECTRIC CURRENT AND ITS EFFECTS — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

📖 Topics in this chapter:

1. Bulb Filament: Material, Properties & Resistance Formula
2. Solenoid: Magnetic Field & Applications
3. Fleming's Left-Hand Rule vs Right-Hand Rule
4. Magnetic Field Lines: Properties & Patterns
5. Electromagnetic Induction: Faraday's Law & Lenz's Law
6. Resistance: Formula, Factors, Resistivity & Temperature
7. Ohm's Law & V-I Graphs
8. Series vs Parallel Circuits
9. DC Motor, DC Generator & AC Generator
10. Force on Conductor
11. Heating Effect of Electric Current (Joule's Law)
12. Ammeter, Voltmeter & Galvanometer
13. Fuse: Principle, Material & Rating
14. Conductors, Insulators, Semiconductors & Superconductors
15. Energy Conversions
16. Coulomb's Law
17. Electric Current Basics
18. Magnetic Compass & Field Detection
19. Oersted's Experiment, Medical Imaging, Electroplating & Solutions
20. Power Formula & Basic Components
21. ⚠️ Traps / Formula Cheat Sheet (MCQ Q405–Q632)
22. Numerical: Ohm's Law ($V = IR$)
23. Numerical: Resistance Formula ($R = \rho l/A$) & Wire Stretching
24. Numerical: Series & Parallel Resistance Combinations
25. Numerical: Electric Power ($P = VI = I^2R = V^2/R$)
26. Numerical: Heat/Energy ($H = I^2Rt = VIt = V^2t/R$)
27. Numerical: Charge & Current ($Q = It$)
28. Numerical: Work Done ($W = QV$)
29. Numerical: Energy Units ($kWh \rightleftharpoons \text{Joules}$)
30. Numerical: Bulb Brightness & Power Ratings
31. Numerical: Cost Calculation & Fuse Rating
32. Numerical: Induced EMF, Coulomb's Law, Magnetic Field & Electrons
33. Numerical: Wire Doubled on Itself
34. Numerical: Temperature Conversion & Cross-Topic Numericals
35. ⚠️ Traps / Master Formula Cheat Sheet (Numerical Q633–Q835)

1. Bulb Filament: Material, Properties & Resistance Formula

- Bulb filament mein Tungsten (W, atomic no. 74) use hota hai kyunki iska melting point sabse zyada ($\approx 3380^\circ\text{C}$ – 3422°C) kisi bhi metal mein. [Q364, Q405, Q418, Q543, Q599]
- Filament ko thin aur long banaya jaata hai taaki resistance zyada ho — $R = \rho l/A$ (lambda length l + chhota cross-sectional area A = zyada R). [Q405]
- High resistance se zyada heat aur light produce hoti hai; temperature zyada = efficiency zyada = light safed (white). [Q405, Q556]
- Filament material ke liye zaroori properties: High melting point, high resistivity, high temperature par glow karne ki ability, ductile nature, easily oxidize na hona. [Q418]

- Bulb mein chemically inactive gases fill hoti hain: Nitrogen (N_2) aur Argon (Ar) — filament ki oxidation aur evaporation prevent karne ke liye. [Q405, Q583, Q606]
- Hydrogen, Oxygen, Chlorine reactive hain → bulb mein use nahi hote. [Q583]
- First commercial electric bulb: Thomas Alva Edison (1879). [Q543, Q599]
- Bulb glow karta hai kyunki filament current se heat hokar high temperature tak pahunchta hai (Joule heating). [Q556]
- Incandescent lamp = heating effect par based hai; electric geyser bhi. [Q417, Q420]

2. Solenoid: Magnetic Field & Applications

- Solenoid = tightly packed helix mein wound coil; electromagnet ki tarah kaam karta hai jab current flow hoti hai. [Q406, Q430, Q433, Q439, Q452, Q502]
- Solenoid bar magnet ki tarah behave karta hai — ek end North pole, doosra South pole. [Q406, Q433]
- Solenoid ka strong magnetic field soft iron ko magnetize karne ke liye use ho sakta hai. [Q406, Q452]
- Solenoid ke andar field lines parallel straight lines hain → magnetic field uniform aur non-zero hota hai, isliye ek uniform field indicate karta hai. [Q406, Q430, Q452, Q463, Q502, Q505, Q576]
- Solenoid ke andar field lines concentric circles nahi hain; ye straight conductor ke liye hoti hain. [Q406]
- Solenoid ka magnetic field strength formula: $B = \mu_0 NI / l = \mu_0 n I$, jahan n = turns per unit length, I = current, l = length. So $B \propto nI$. [Q439, Q463, Q488, Q490]
- B depends on: Current (I), Number of turns (N/n), Length (l inversely), Core material permeability. Circular coil ka field bhi turns (n) ke directly proportional hota hai — jitne zyada turns, utna strong field. [Q439, Q446, Q474, Q488, Q490, Q503]
- Soft iron core ka use: magnetic permeability bahut high hoti hai → field greatly amplify hota hai; length of metallic core aur current bhi electromagnet ki strength affect karte hain — coil ka placement direction affect nahi karta. [Q472, Q474, Q496]
- Wood, glass, ebonite = non-magnetic → field amplify nahi karte. [Q496, Q503]
- Direction change (current reverse) → sirf direction change hoti hai, magnitude nahi. [Q474]
- Constant current I → constant uniform magnetic field B inside solenoid. [Q490]
- Loudspeaker, Voltmeter, Ammeter = force on current-carrying conductor principle par based. [Q452]
- Circular coil mein 6 turns = single loop se 6 times zyada magnetic field; kyunki har turn ka field add hota hai (same direction mein). [Q414]

3. Fleming's Left-Hand Rule vs Right-Hand Rule

- Fleming's Right-Hand Rule = Generator/Induced current ke liye. [Q408, Q426, Q428, Q451, Q498, Q537]
 - Forefinger → Magnetic field direction
 - Middle finger → Induced current/EMF direction
 - Thumb → Motion of conductor (force)
 - Teenon mutually perpendicular hote hain. [Q498]
 - Sirf tab valid jab conductor magnetic field ke perpendicular (90°) move kare. [Q428, Q476, Q534]
 - Example: horseshoe magnet North west, South east, rod vertically downward move ho → induced current north to south direction mein flow karega. [Q537]
- Fleming's Left-Hand Rule = Motor/Force on conductor ke liye. [Q408, Q416, Q429, Q431, Q454, Q473, Q481, Q501, Q518, Q521, Q522, Q529]
 - Forefinger → Magnetic field direction
 - Middle finger → Current direction

- Thumb → Force/Motion direction
- Example: field north-to-south (downward), wire deflected south → conventional current west to east, isliye electrons (opposite direction) east to west flow karte hain. [Q473]
- Right-Hand Thumb Rule (Maxwell's Corkscrew Rule) = current-carrying conductor ke around magnetic field ki direction nikaalne ke liye. [Q408, Q415, Q422]
 - Thumb → current direction; Curled fingers → magnetic field direction.
- Conductor par force ki direction tab badalti hai jab current ki direction reverse ki jaaye ya magnetic field ki direction reverse ki jaaye. [Q431, Q481, Q522]
- Current ya magnetic field ka magnitude change karne se force ka magnitude badalta hai, direction nahi. [Q431, Q522]
- Motion aur induced EMF ki direction perpendicular hoti hain; same ya opposite nahi. [Q498]

4. Magnetic Field Lines: Properties & Patterns

- Magnetic field lines kabhi intersect nahi karti hain — agar karti to ek point par do directions ho jaati, jo impossible hai. [Q409, Q410, Q432, Q434, Q497]
- Field lines hamesha closed continuous loops banati hain. [Q409, Q410, Q432]
- Bar magnet ke bahar: field lines North se South ki taraf jaati hain; andar: South se North. [Q409, Q410, Q432, Q434, Q456, Q485]
- North pole se emerge hoti hain, South pole par merge hoti hain. [Q456, Q465]
- Field lines ki density/crowding = field strength; poles par sabse dense (strongest), center ya beech mein kam. [Q410, Q434, Q466, Q492, Q509]
- Uniform magnetic field mein field lines parallel aur equidistant hoti hain. [Q409]
- Current-carrying straight conductor ke around field lines concentric circles hoti hain (hemispherical, intersecting ya parallel nahi), radius conductor se doori ke saath badhta hai (strength kam hoti hai). [Q409, Q422, Q444, Q464, Q467, Q520, Q532]
- Conductor se door jaane par magnetic field strength kam hoti hai — inverse relation ($B \propto 1/r$); current zyada hone par field strength badhti hai ($B \propto I$). [Q422, Q464, Q509, Q520]
 - Points P(r), Q(2r), S(3r) → $B_P > B_Q > B_S$. [Q464]
- Current ki direction reverse karne par field lines ki direction bhi reverse ho jaati hai. [Q440]
- Wire ko East direction mein current le jaaye to Right-Hand Thumb Rule se field wire ke neeche South direction mein banega. [Q514]
- Compass needle ka North pole field ki direction batata hai, South pole nahi. [Q440]
- Field lines ki closeness = relative strength of magnetic field. [Q466]
- Magnetic field lines have both magnitude and direction — vector quantity hain. [Q466, Q497]
- Magnet ki strength $\propto$ number of field lines. Agar A se 5 lines aur B se 10 lines → A ki strength half of B. [Q465]
- Field ki direction kisi bhi point par tangent se milti hai. [Q512]

5. Electromagnetic Induction: Faraday's Law & Lenz's Law

- Electromagnetic Induction = changing magnetic flux ke kaaran coil mein EMF/induced current generate hona. Discovery: Michael Faraday. [Q411, Q419, Q424, Q425, Q441, Q442, Q449, Q453, Q482, Q493, Q506, Q524, Q545, Q575, Q618]
- Electric generator mein mechanical energy use hoti hai conductor ko magnetic field mein rotate karne ke liye electricity produce karne ke liye. [Q449]
- Faraday's Law: Induced EMF = $N \times (d\Phi/dt)$ = rate of change of magnetic flux ke proportional. [Q419, Q442, Q453, Q482, Q493, Q506, Q524]

- Faraday's Second Law: Induced EMF = rate of change of magnetic flux se proportional ($EMF = N \times d\Phi/dt$). [Q453]
- Sirf flux change hone par induction hoti hai — flux itself nahi, uska rate of change matter karta hai. [Q453]
- EMF independent of R (resistance); depends on: magnetic strength, turns, cross-sectional area, flux change rate. [Q506]
- Induced current $I = EMF/R \rightarrow$ depends on R. [Q506]
- Induced current tabhi hoti hai jab relative motion ho magnet aur coil ke beech:
 - Magnet stationary rakhne par $\rightarrow$ flux change nahi $\rightarrow$ no induced current. [Q424, Q441, Q450, Q458, Q524]
 - Magnet ko towards ya away coil se move karne par $\rightarrow$ flux change $\rightarrow$ induced current. [Q424, Q539]
 - Coil aur magnet same direction mein same speed se move karein $\rightarrow$ relative motion zero $\rightarrow$ no induction. [Q442, Q457, Q539]
 - Magnet ko solenoid ki axis par rotate karne par $\rightarrow$ flux change nahi $\rightarrow$ zero induced current. [Q507]
- Square coil rotating magnetic field mein ho to induced current ki direction har half revolution mein change hoti hai (AC generation ka principle). [Q515]
- Key switch on/off karne ke interval mein induction hoti hai; steady current hone par nahi. [Q441, Q450, Q458]
 - Key closed karne ke waqt: current 0 se max tak badhti hai $\rightarrow$ flux change $\rightarrow$ induction hoti hai. [Q450]
 - Key closed hone ke 2 minutes baad: current steady $\rightarrow$ flux constant $\rightarrow$ no induction. [Q450, Q458]
 - Current decrease at constant rate (rheostat se) $\rightarrow$ flux change $\rightarrow$ induction hoti hai. [Q458]
- Induced potential difference (EMF) ko increase karne ke liye:
 - Coil mein turns (N) zyada karo. [Q438]
 - Stronger magnet (zyada magnetic field) use karo. [Q438]
 - Magnet ko tez speed se move karo. [Q419, Q442, Q460]
- Maximum induced EMF jab coil aur magnet ek doosre ki taraf move karein (relative velocity = $v_1 + v_2$). [Q442]
- Coil ki speed kam hone par $\rightarrow$ flux change rate kam $\rightarrow$ induced EMF/potential difference kam hota hai. [Q482]
- AC generator ke armature ko tez rotate karne par $\rightarrow$ flux change rate zyada $\rightarrow$ induced potential difference (EMF) increases. [Q460]
- Lenz's Law: Induced current ki direction hamesha us cause ka oppose karti hai jisne use produce kiya. [Q483, Q493]
- Magnet ka North pole coil ki taraf $\rightarrow$ induced current ek direction mein. [Q483]
- Same magnet, same speed, South pole coil ki taraf $\rightarrow$ induced EMF ka magnitude same, lekin sign/direction opposite. [Q483]
- Straight conductor ki current direction continuously change hone par $\rightarrow$ magnetic field continuously change $\rightarrow$ induced current not constant (continuously changing). [Q493]
- Induced current maximum jab conductor ki motion magnetic field ke perpendicular (90°) ho. [Q476]
 - 0° par minimum (zero) induced current. [Q476, Q534]
 - $EMF = BLv \sin\theta$; $\theta = 0^\circ \rightarrow \sin 0^\circ = 0 \rightarrow EMF = 0$. [Q534]

6. Resistance: Formula, Factors, Resistivity & Temperature

- Resistance formula: $R = \rho l/A$; ρ = resistivity, l = length, $A = \pi r^2$. [Q405, Q412, Q413, Q427, Q478, Q487, Q510, Q513, Q589, Q602, Q620]
- $R \propto l$ (length):
 - Length double $\rightarrow$ R double. [Q510]
 - Length 5 times $\rightarrow$ R 5 times. [Q478]
 - Length double karke, area same rakhne par ammeter reading (current) half ho jaati hai V constant hone par (kyunki R double hone se $I = V/R$ half ho jaata). [Q630]
- $R \propto 1/A$ (area):

- Area double $\rightarrow R$ half. [Q602, Q620]
- Area half karne par R double. [Q487 context]
- Length double + Area half karne par $\rightarrow R = \rho(2l)/(A/2) = 4$ times original. [Q487]
- Resistivity (ρ) = material ki intrinsic property; length ya area par depend nahi karti. [Q412, Q427, Q487, Q553, Q601]
 - Resistivity sirf material nature aur temperature par depend karti hai. [Q412, Q423, Q427, Q487]
 - Wire ko 4 parts mein kaatne par bhi ρ same rehti hai. [Q412]
 - SI unit = Ohm-meter ($\Omega \cdot m$). [Q553, Q587]
 - Definition: 1m length aur $1m^2$ cross-section wale conductor ka resistance. [Q553]
- Temperature badhne par pure metals ka resistance badhta hai (electron-ion collisions zyada hote hain). [Q423, Q560]
- Semiconductors ka resistance temperature ke saath kam hota hai. [Q617]
- Resistance does NOT depend on: density (Q580), pressure (Q581). [Q580, Q581]
- Resistivity order (low to high):
 - Silver (1.60×10^{-8}) < Copper (1.62×10^{-8}) < Aluminium (2.63×10^{-8}) < Tungsten (5.20×10^{-8}) < Manganin (44×10^{-6}) < Glass/Rubber (10^{12} – 10^{17}). [Q443, Q477, Q479, Q491, Q528, Q587, Q624]
- Same dimension mein: Silver ka resistance sabse kam hoga (lowest ρ). [Q479, Q491, Q587]
- Alloys ki resistivity generally pure metals se zyada hoti hai. [Q427]
- Copper transmission lines ke liye ideal hai kyunki iska resistance kam hai $\rightarrow$ energy loss kam. [Q436]
 - Aluminium bhi use hota hai kyunki lightweight aur cheap hai. [Q567]
- High resistivity aur low melting point transmission line material ke liye undesirable hain. [Q436]
- Nichrome = metals mein maximum resistivity; isliye heating elements mein use hota hai. [Q443, Q592]
 - Properties: high resistance, high melting point, oxidation resistant. [Q592]
- Resistors = alloys se bane hote hain (Nichrome, Constantan, Manganin) $\rightarrow$ high resistivity. [Q570]
- Resistivity order category-wise: Conductors < Alloys < Insulators. [Q528]

7. Ohm's Law & V-I Graphs

- Ohm's Law (1827): $V = IR \rightarrow$ current (I) voltage (V) ke directly proportional hai, jabki resistance (R) constant ho. [Q447, Q486, Q489, Q508, Q519, Q540, Q569, Q588, Q594, Q621]
- V-I graph = straight inclined line through origin $\rightarrow$ linear. [Q448, Q486, Q508, Q594]
 - Slope = Resistance (R). [Q486]
 - Steeper slope $\rightarrow$ zyada current for same voltage $\rightarrow$ lower resistance. [Q407]
- Do alag materials ka V-I graph dono straight line through origin $\rightarrow$ dono Ohm's law follow karte hain; sirf slope alag hai $\rightarrow$ resistance alag hai. [Q448]
- Constant temperature par V increase karein $\rightarrow I$ proportionally increase karegi (R constant rehta hai). [Q489]
- Temperature badhne par R badhta hai $\rightarrow$ current aur drift velocity dono kam ho jaate hain. [Q489]
- Drift velocity formula: $V_d = I/(nAe)$; n = electron density, A = area, e = electron charge. [Q489]
- Resistance unit: Volt/Ampere = Ohm (Ω). [Q519, Q526]
- Current $\propto 1/R$ (inverse). [Q407]
- Agar V constant aur I increase $\rightarrow R$ decrease ($R = V/I$). [Q621]
- Agar R $4\times$ ho jaye (V same) $\rightarrow$ current $I/4$. [Q447]

8. Series vs Parallel Circuits

- Electrical circuit connections do types ki hoti hain: Series aur Parallel. [Q614]
- Series connection: components end-to-end; single path for current. [Q533, Q554, Q565, Q593, Q616]
 - Same current through all components (koi bhi resistor ho, current same rehta hai). [Q475, Q500, Q461, Q593]
 - Total resistance: $R_s = R_1 + R_2 + R_3 + \dots$ [Q461, Q616]
 - Voltage divides: $V_{\text{total}} = V_1 + V_2 + V_3$. [Q540]
 - Ek bulb blow $\rightarrow$ sabhi bulbs stop glowing (circuit breaks). [Q565]
 - Examples: Decoration lights (Diwali ki jhalar), water heater, freezer, refrigerator, lamp, flashlight. [Q533]
- Parallel connection: components across same two points; multiple paths. [Q552, Q555, Q616]
 - Same voltage across all components. [Q555, Q619]
 - Total resistance: $1/R_p = 1/R_1 + 1/R_2 + 1/R_3 + \dots$ [Q435]
 - Current divides; total current = sum of branch currents. [Q500]
 - Ek component fail $\rightarrow$ others continue to work. [Q565]
 - Examples: Home wiring, street lights, car headlights, DC filter circuit, fire alarm. [Q533, Q555]
- Domestic wiring parallel mein hoti hai taaki devices independently kaam karein. [Q555]
- Parallel mein jude bulbs zyada bright glow karte hain series ke comparison mein. [Q500]
- Series mein potential difference across each resistor uske resistance ke proportional hota hai ($V = IR$, I same $\rightarrow V \propto R$). [Q455]
- Parallel mein 2 resistors: $R = R_1 R_2 / (R_1 + R_2)$. [Q435]
- 4 wires same material, garland (mala) ki tarah series mein $\rightarrow R_{\text{eq}} = 4R \rightarrow 4$ times original. [Q461]
- P point (before split) par current Q point (after split) se zyada hoti hai. [Q500]

9. DC Motor, DC Generator & AC Generator

- DC Motor = electrical energy $\rightarrow$ mechanical energy convert karta hai. [Q416, Q469, Q471, Q551, Q608, Q612]
 - Working principle: Current-carrying conductor magnetic field mein rakha jaaye to mechanical force lagti hai. [Q416]
 - Split rings (commutator) har half rotation par current ki direction reverse karte hain $\rightarrow$ continuous rotation possible hoti hai. [Q416, Q471, Q495, Q538]
 - Maximum force tab hoti hai jab coil ke arms magnetic field ke perpendicular hote hain; parallel hone par force kam. [Q416]
 - Force ki direction: Fleming's Left-Hand Rule se determine hoti hai. [Q416]
 - Motor elements: Armature Core (coil hold), Commutator (current reverse), Carbon Brushes (current conduct), Magnetic Field (torque). [Q538]
 - Commercial motor features: Electromagnet use karta hai (permanent magnet ki jagah) $\rightarrow$ zyada torque; Soft-iron core $\rightarrow$ current band hone par magnetism retain nahi karta. [Q470]
 - Motor invention: Michael Faraday (1821). [Q471, Q551]
 - Motor uses: Fan, Mixer, Washing machine, Computer, MP3 player, Refrigerator, Pump, Vacuum cleaner. [Q462, Q551]
 - DC motor types: Series, Shunt/Parallel, Compound, Permanent Magnet DC motor. [Q416]
 - Galvanometer motor nahi use karta — ye sirf current detect/measure karta hai; iron core + rectangular wire loop se bana hota hai. [Q462]
- DC Generator (Dynamo) = mechanical energy $\rightarrow$ electrical energy (DC). [Q451, Q495, Q504, Q575, Q618]
 - Uses split-ring commutator to reverse current every half rotation. [Q495, Q504]
 - Current flows in one direction only. [Q504]

- Generates lower voltages. [Q504]
- Current induced only in rotor. [Q504]
- Principle: Electromagnetic Induction (Faraday's Law). [Q545, Q575, Q618]
- AC Generator (Alternator) = mechanical energy → electrical energy (AC). [Q451, Q504, Q618]
 - Converts mechanical → electrical (AC).
 - Uses slip rings (not commutator). [Q495, Q504]
 - Current direction periodically reverses.
 - Can generate very high voltages.
 - Current can be induced in rotor or stator.
 - Uses: Power stations, large-scale electricity generation. [Q618]
- Commutator use hota hai: DC motor aur DC generator dono mein. [Q495]
 - AC generator mein nahi. [Q495]
- Transformer = transfers electrical energy from one circuit to another through electromagnetic induction. [Q579]

10. Force on Conductor

- Force formula: $F = BIL \sin\theta$. [Q494, Q518]
- Force maximum jab $\theta = 90^\circ$ (arms perpendicular to magnetic field) → $\sin 90^\circ = 1 \rightarrow F_{\text{max}} = BIL$. [Q494]
- Force zero jab $\theta = 0^\circ$ (parallel to field) → $\sin 0^\circ = 0$. [Q494, Q518]
- Force ki direction Current ki direction aur magnetic field ki direction dono par depend karti hai. [Q445]
- Force ki direction: Fleming's Left-Hand Rule se. [Q416, Q431, Q445, Q454, Q481, Q501, Q521, Q529]
- Magnetic field (North), Current (East to West) → Force direction: Downward (Fleming's Left-Hand Rule se). [Q454]
- Force ki direction tab badalti hai jab current ki direction reverse ki jaaye ya magnetic field ki direction reverse ki jaaye. [Q431, Q481, Q522]
- Current ya magnetic field ka magnitude change karne se force ka magnitude badalta hai, direction nahi. [Q431, Q522]
- Motor mein coil ki do arms mein current opposite directions mein flow hoti hai → forces equal magnitude, opposite direction → rotational torque (couple) create hota hai. [Q521, Q529]

11. Heating Effect of Electric Current (Joule's Law)

- Formula: $H = I^2RT$ — heat produced directly proportional to square of current, resistance, aur time. [Q363, Q371, Q374, Q417, Q420, Q535, Q544, Q591]
- Heating effect par based devices: Electric iron, Electric heater, Electric geyser, Induction cooker, Microwave oven, Hair blower, Incandescent lamp, Immersion rod, Electric kettle, Electric toaster, Electric oven, Fuse. [Q363, Q365, Q371, Q372, Q417, Q420, Q517, Q547, Q579, Q591]
- NOT based on heating effect:
 - Electric generator → electromagnetic induction par. [Q417, Q575]
 - Electric washing machine, mixer, fan, computer, refrigerator → electric motor use karte hain. [Q420, Q462, Q547, Q579]
 - Electric bell, electromagnet → magnetic effect par. [Q363]
 - Speaker → electromagnetic induction par. [Q363]
 - Transformer → induction par. [Q579]
- Electric current ke teen main effects hote hain: Heating, Magnetic aur Chemical — "Contraction effect" naam ka

koi recognized effect nahi hai. [Q598]

- Electric iron ka invention: Henry Seely White (1882). [Q420]
- Electric circuits mein losses heat ke form mein hoti hain (Joule's heating). [Q535]
- Factors: Resistance (zyada R = zyada heat), Time (zyada t = zyada heat), Current (zyada I = zyada heat).
 - Current triple ho jaye (R , t constant) $\rightarrow$ heat $H \propto I^2$ se 9 times ho jaati hai. [Q468, Q535]
 - Resistance half ho jaye (I , t constant) $\rightarrow$ heat bhi half ho jaati hai ($H \propto R$). [Q628]
 - Energy dissipated formula alternate loop mein: $H = I^2Rt = IVt = VIt$ (Ohm's law $V=IR$ use karke). [Q629]
- Brightness vs Resistance: Same voltage source par bulb with higher resistance $\rightarrow$ less current ($I = V/R$) $\rightarrow$ less power ($P = V^2/R$) $\rightarrow$ dimmer. [Q586]
 - So dim bulb = higher resistance; bright bulb = lower resistance. [Q586]

12. Ammeter, Voltmeter & Galvanometer

- Ammeter = current measure karta hai; series mein lagta hai; very low resistance (ideally zero) taaki current significantly reduce na ho. [Q557, Q562, Q590, Q625]
- Voltmeter = potential difference measure karta hai; parallel mein lagta hai; very high resistance (ideally infinite) taaki negligible current le. [Q562, Q571, Q590, Q596, Q610]
- Galvanometer = very small current (≈ 100 mA) ki presence aur direction detect karta hai. [Q462, Q585, Q625]
 - Iron core + rectangular wire loop se bana hota hai.
 - Motor nahi use karta. [Q462]
- Galvanometer ko voltmeter banane ke liye high resistance series mein lagate hain. [Q462]
- Rheostat current control karta hai; Ohm's Law par based. [Q568, Q600, Q605]
- Resistance symbol: zigzag line (American) ya rectangle (European). [Q526]
- Electric bulb symbol = circle with X inside. [Q626]
- Voltmeter symbol = circle with V and $\pm$ terminals. [Q610]
- Rectifier = AC ko DC mein convert karta hai. [Q549]

13. Fuse: Principle, Material & Rating

- Fuse = safety device based on heating effect of current ($H = I^2Rt$). [Q544, Q591]
- Fuse wire = Tin (Sn) + Lead (Pb) alloy; low melting point hota hai. [Q544, Q613]
- Jab current rated value se zyada ho jaaye $\rightarrow$ wire heat hokar melt ho jaata hai $\rightarrow$ circuit break $\rightarrow$ protection. [Q544, Q613]
- Fuse rating example: 8 A $\rightarrow$ 8 A tak safe, usse zyada par melt. [Q613]
- Domestic fuse rating: 1 Ampere – 10 Ampere. [Q536]
 - Small appliances (bulb, fan): ~ 5 A.
 - Heavy appliances (AC, geyser): up to 15A.
- Fuse wire properties: Low melting point, high conductivity, low oxidation. [Q459]
- Fuse series mein connected hota hai device ke saath. [Q459, Q590]
- Short circuit = low-resistance connection between conductors $\rightarrow V = IR$, $R \approx 0 \rightarrow I = V/R$ becomes enormously large. [Q511, Q584]
 - Causes: Vermin chewing wires, loose connections, old/damaged outlets. [Q511]
- PVC coating on wires $\rightarrow$ insulation $\rightarrow$ short circuit prevent karta hai. [Q584]

- Electrician rubber gloves pehanta hai → rubber = excellent insulator → electric shock se bachata hai. [Q558]

14. Conductors, Insulators, Semiconductors & Superconductors

- Conductors = low resistivity; allow current flow easily. [Q480, Q564, Q587, Q617]
 - Examples: Silver (lowest $\rho = 1.60 \times 10^{-8} \Omega\text{m}$), Copper (1.62×10^{-8}), Aluminium (2.63×10^{-8}), Gold, Iron. Human body bhi conductor hai kyunki body cells mein ions hote hain jo electricity conduct karte hain. [Q477, Q479, Q491, Q546, Q587, Q567]
 - Copper aur Aluminium = most common for transmission wires (copper: low ρ , aluminium: lightweight + cheap). [Q436, Q567]
- Insulators = extremely high resistivity (practically infinite). [Q480, Q564, Q595, Q611]
 - Resistivity order: $10^{12} \Omega\text{m}$ to $10^{17} \Omega\text{m}$. [Q624]
 - Examples: Glass, Rubber, Plastic, Wood, Dry air, Bakelite, Ebonite, Paper, Cotton, Diamond, Mica, Quartz. [Q443, Q477, Q564, Q573, Q595, Q604, Q611, Q631]
 - Ebonite resistivity = 10^{15} to $10^{17} \Omega\text{m}$ → highest among common options. [Q604]
 - Bakelite = polymer (phenol + formaldehyde) → thermosetting insulator → weakest conductor. [Q573]
- Semiconductors = conductors aur insulators ke beech. [Q570, Q617]
 - Examples: Silicon (Si), Germanium (Ge).
 - Resistivity decreases with increasing temperature.
- Superconductors = zero resistance. Examples: Magnesium Diboride, Niobium. [Q564]
- At high temperatures, conductor resistivity increases and conductivity decreases, while for semiconductors, resistance decreases with increasing temperature. [Q617]
- Good conductor (transmission lines) ke liye chahiye: low resistance, free electron movement. [Q436]
- High resistivity aur low melting point transmission line material ke liye undesirable hain. [Q436]
- Nichrome = heating devices mein use hota hai (water heater coils, hair dryer, toaster, oven) → high resistance + high melting point + oxidation resistant. [Q443, Q592]

15. Energy Conversions

- Electric Motor = Electrical energy → Mechanical energy. [Q416, Q469, Q471, Q551, Q608, Q612]
- Electric Generator/Dynamo = Mechanical energy → Electrical energy. [Q417, Q449, Q451, Q469, Q575, Q618]
- Heating Effect Devices = Electrical → Heat energy. [Q363, Q371, Q417, Q420, Q517, Q547, Q579, Q591]
- Electric bulb = Electrical → Light + Heat. [Q612]
- Microphone = Sound → Electrical. Loudspeaker = Electrical → Sound. [Q612]
- Battery/Cell = Chemical energy → Electrical energy. [Q551, Q572, Q603]
 - Primary cell = non-rechargeable (Leclanche, Daniel, Dry Cell). [Q603]
 - Invented by Alessandro Volta. [Q559, Q603]
- Photocell = Light energy → Electrical energy. [Q551]
- Torch = Chemical (cells) → Electrical → Light. [Q608]
- TV = Electrical → Light + Sound. [Q608]
- Electromagnet = Electrical energy → Magnetic energy. [Q484]
- Turbine = Hydraulic energy → Mechanical energy. [Q469]
- Transformer = transfers electrical energy from one circuit to another through electromagnetic induction. [Q579]

16. Coulomb's Law

- Coulomb's Law: Do point charges ke beech attraction/repulsion force directly proportional to product of charges aur inversely proportional to square of distance. [Q622, Q623]
- Formula: $F = k \times (q_1 q_2 / r^2)$; $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$. [Q622, Q623]
- Scientist: Charles Coulomb. [Q622]

17. Electric Current Basics

- Electric current = flow of electrons (negative charges) in conductor. [Q607, Q627]
- Conventionally, current direction = positive charges ka flow (positive to negative terminal). [Q577, Q582]
 - Actual electron flow = negative to positive (opposite). [Q582]
- Current flow ke liye potential difference zaroori hai. Agar do points ka potential equal ho $\rightarrow$ no current flows. [Q597]
- Electric potential kisi point par = work done in moving a unit positive charge infinity se us point tak (bina acceleration ke). [Q566]
- Potential difference (V) do points ke beech = work done per unit charge ek point se doosre point tak move karne mein: $V = W/Q$. [Q578]
- SI unit = Ampere (A) = 1 Coulomb/second. [Q607]
- Electric circuit = continuous closed path for current flow. [Q561, Q616]
- Cell/battery maintains potential difference $\rightarrow$ drives continuous current. [Q572]
- Conductor mein current flow = electrons ka motion; solution mein = ions ka motion (electrolyte). [Q627]
- Neutrons = neutral (no charge), protons = fixed in nucleus $\rightarrow$ dono current carry nahi karte. [Q627]

18. Magnetic Compass & Field Detection

- Magnetic compass = tiny magnetic needle (aluminium pointer se attached) jo North-South direction indicate karti hai. [Q499, Q527]
 - China mein invented (2nd century BC - 1st century AD). [Q499]
- Compass needle bar magnet ke paas laane par deflect hoti hai kyunki ye khud ek chhota magnet hai. [Q527]
- Magnetic field ki presence detect karne ke liye magnetic needle use hoti hai; galvanometer, voltmeter, ammeter nahi. [Q530]
- Freely suspended bar magnet hamesha geographical North-South direction mein point karta hai. [Q421, Q434]
 - Reason: Earth ka magnetic South pole geographical North mein hai (unlike poles attract). [Q421]
- Magnetic field = magnet ke around ka region jahan uska effect detect ho sakta hai. Unit = Tesla. [Q615]
- Electromagnetic force = electrically charged particles ke interaction se. [Q609]
 - Friction, tension, spring force = all electromagnetic at microscopic level.
 - Weight (gravitational force) = NOT electromagnetic; alag fundamental force hai. [Q609]

19. Oersted's Experiment, Medical Imaging, Electroplating & Solutions

- Oersted's Experiment (1820): Current-carrying wire ke paas compass needle deflect hoti hai $\rightarrow$ wire magnetic field produce karti hai. [Q531]
 - Current direction reverse karne par deflection ki direction bhi reverse hoti hai. [Q531]

- Medical Imaging: [Q523]
 - MRI = Powerful magnetic field + radiofrequency pulses + computer.
 - CT Scan = X-rays + computer → cross-sectional images.
 - Ultrasound = High-frequency sound waves.
 - PET Scan = Radioactive substance (tracer).
- Electroplating = electricity se kisi material par desired metal ki layer deposit karna. [Q548]
- Electrolytes = solutions jo electricity conduct karte hain (acids, bases, salts in water). [Q550]
- Non-electrolytes = alcohol, glucose → no free ions → do NOT conduct electricity. [Q550]
- Distilled water = poor conductor (no dissolved salts/ions); regular tap water conducts kyunki dissolved salts hain. [Q574]
- Parallel conductors: same direction current → attract; opposite direction → repel. [Q525]
- GK crossover facts jo isi chapter mein aaye: sabse effective tarika electricity production kam karne ka hai combustion (fuel burning) kam karna, kyunki thermal power plants combustion se hi electricity banate hain (reduce/reuse/composting alternative waste-management concepts hain, production kam karne ka direct tarika nahi). [Q516]
- Water ki density sabse zyada 277 K (4°C) par hoti hai — 273 K (0°C, ice) par nahi; heating se molecular cage-structure break hoke volume minimum aur density maximum ho jaati hai. [Q542]

20. Power Formula & Basic Components

- Electric power formulas: $P = VI = I^2R = V^2/R$. [Q563]
 - IR^2 power represent nahi karta → wrong dimensions. [Q563]
- Resistor = passive two-terminal component jo electrical resistance implement karta hai; current ko limit/regulate karta hai. [Q541, Q526]
- Capacitor = electrical energy store karta hai electric field mein. [Q541]
- Transistor = miniature semiconductor jo current/voltage regulate aur amplify karta hai. [Q541]
- Diode = one-way switch for current. [Q541]
- Switch = circuit ko break/close karne ke liye; invented by John Henry Holmes (1884). [Q559]
- Cell/battery: Invented by Alessandro Volta. [Q559, Q603]

21. ⚠ Traps / Formula Cheat Sheet (MCQ Q405–Q632)

- ⚠ Trap: Fleming's Right = Generator (induced current); Left = Motor (force). Ulta mat karna! [Q408, Q416, Q426, Q451, Q498]
- ⚠ Trap: Bulb mein Nitrogen aur Argon fill hoti hai; Hydrogen/Oxygen nahi. [Q583, Q606]
- ⚠ Trap: Filament = Tungsten; Silver/Iron/Nickel nahi. [Q543, Q599]
- ⚠ Trap: Solenoid ke andar field lines parallel straight hain; straight conductor ke around concentric circles. [Q406, Q422, Q467]
- ⚠ Trap: Magnetic field strength poles par strongest hoti hai, center par nahi. [Q410]
- ⚠ Trap: Resistivity (ρ) length/area par depend nahi karti — sirf material aur temperature par. [Q412, Q427, Q487, Q601]
- ⚠ Trap: Length double + Area half = R 4 times (not 2 times or unchanged). [Q487]
- ⚠ Trap: Resistance density ya pressure par depend nahi karti. [Q580, Q581]
- ⚠ Trap: Temperature badhne par pure metals ka R badhta hai; semiconductors ka R kam hota hai. [Q423, Q617]
- ⚠ Trap: Induced EMF flux change se nahi, rate of change of flux se proportional hai. [Q453, Q482]

- ⚠ Trap: Same direction same speed mein coil aur magnet move karein → no induction (relative motion zero). [Q457, Q539]
- ⚠ Trap: Magnet North pole vs South pole same speed par → EMF magnitude same, direction opposite. [Q483]
- ⚠ Trap: Commutator DC motor + DC generator dono mein use hota hai; AC generator mein slip rings. [Q495, Q504]
- ⚠ Trap: Force maximum at 90° (perpendicular), zero at 0° (parallel). [Q494, Q518, Q534]
- ⚠ Trap: Force direction sirf current direction ya field direction change karne se badalti hai; magnitude change se nahi. [Q522]
- ⚠ Trap: Series mein current same, voltage divide; Parallel mein voltage same, current divide. [Q475, Q500, Q533, Q540, Q593, Q619]
- ⚠ Trap: EMF R par depend nahi karti; induced current depend karti hai ($I = EMF/R$). [Q506]
- ⚠ Trap: Domestic wiring parallel mein; series nahi. [Q555]
- ⚠ Trap: Ammeter = series + low R; Voltmeter = parallel + high R. [Q557, Q571, Q590]
- ⚠ Trap: Motor = electrical → mechanical; Generator = mechanical → electrical. Ulta mat karo! [Q612, Q618]
- ⚠ Trap: Toaster = heating effect; motor nahi use karta. [Q579]
- ⚠ Trap: Equal potential par no current flows; potential difference zaroori hai. [Q597]
- ⚠ Trap: IR^2 power represent nahi karta; I^2R karta hai. [Q563]
- ⚠ Trap: Cell = chemical energy convert karta hai; potential/kinetic nahi. [Q603]
- ⚠ Trap: Weight = gravitational force; electromagnetic nahi. [Q609]
- ⚠ Trap: Compass needle khud chhota magnet hai; sirf conductor nahi. [Q527]
- ⚠ Trap: Oersted experiment mein current reverse → deflection reverse hota hai. [Q531]
- ⚠ Trap: Domestic fuse 1-10A range mein; 100-200A nahi. [Q536]
- ⚠ Trap: Silver = lowest resistivity; Nichrome = high resistivity. [Q587]
- ⚠ Trap: Magnetic field lines vector hain → magnitude + direction. [Q466, Q497]
- ⚠ Trap: Conductor ko field ke parallel move karne par $EMF = 0$ (flux change nahi). [Q534]
- ⚠ Trap: Solenoid mein wooden core field nahi badhata; soft iron badhata hai. [Q472, Q503, Q496]
- ⚠ Trap: Series circuit examples = decoration lights; Parallel = home wiring, street lights. [Q533]
- ⚠ Trap: Repulsion = sure test of magnetism; attraction alone sufficient nahi. [Q432, Q437]
- ⚠ Trap: Earth ka magnetic South pole geographical North mein hai. [Q421]
- ⚠ Trap: Fuse = heating effect par based; electromagnetic induction nahi. [Q591]
- ⚠ Trap: Alloys ki resistivity pure metals se zyada hoti hai. [Q427]
- ⚠ Trap: Glass = insulator = highest resistivity; metals = conductors = low resistivity. [Q443, Q477]
- ⚠ Trap: Fuse wire melts → circuit disconnect; NOT short circuit ya overloading se bachata hai directly. [Q544, Q613]
- ⚠ Trap: Conventional current = positive to negative; actual electrons = negative to positive. [Q577, Q582]
- ⚠ Trap: Magnetic monopoles exist nahi karte. [Q432]

22. Numerical: Ohm's Law ($V = IR$)

- $V = IR$ — sabse basic formula sabhi numerical ka base. [Q633, Q637, Q646, Q647, Q648, Q667, Q672, Q678, Q682, Q683, Q690, Q732, Q745, Q746, Q748, Q757, Q760, Q778, Q779, Q780, Q781, Q783, Q793, Q795, Q796, Q812, Q820]
- Current (I) = V/R — jab voltage aur resistance given ho. [Q633, Q637, Q648, Q706, Q760, Q793, Q795, Q796]
- Resistance (R) = V/I — jab voltage aur current given ho. [Q647, Q672, Q683, Q778, Q779, Q780, Q781, Q783, Q820]
- Voltage (V) = IR — jab current aur resistance given ho. [Q646, Q678, Q682, Q690, Q812]
- V double, R constant → I doubles. [Q748]

- V half, R constant $\rightarrow$ I halves. [Q746]
- V aur I dono double $\rightarrow$ R same rehta hai ($R = V/I$). [Q791]
- Unit consistency zaroori: mA ko A mein convert karo ($1 \text{ mA} = 10^{-3} \text{ A}$). [Q646, Q667]
- V-I graph ka slope = Resistance (jab V vertical, I horizontal). [Q700]

23. Numerical: Resistance Formula ($R = \rho l/A$) & Wire Stretching

- $R = \rho(l/A)$ — ρ = resistivity, l = length, $A = \pi r^2$. [Q634, Q636, Q641, Q651, Q656, Q658, Q663, Q664, Q666, Q677, Q679, Q680, Q698, Q705, Q713, Q714, Q715, Q719, Q721, Q722, Q724, Q734, Q735, Q740, Q751, Q763, Q772]
- Same material (ρ constant) mein:
 - $R \propto l$ — length double $\rightarrow$ R double; length $3\times \rightarrow R 3\times$. [Q478, Q510, Q641, Q705, Q751, Q772]
 - $R \propto 1/A$ — area double $\rightarrow$ R half. [Q656, Q714, Q763]
 - $R \propto 1/r^2$ — radius half $\rightarrow R 4\times$; radius $1/3 \rightarrow R 9\times$. [Q636, Q679]
- Length double + Area half $\rightarrow R = 4\times$ original. [Q487, Q658, Q736]
- Length $4\times$ + Area $5\times \rightarrow R_{\text{new}} = (4/5)\times R$. [Q666]
- Wire stretched (volume constant): $R \propto l^2$ kyunki $A = V/l$. [Q673]
 - Length 50% increase ($1.5\times$) $\rightarrow R_{\text{new}} = (1.5)^2 \times R = 2.25\times R = 9R/4$. [Q673]
- Resistivity (ρ) = RA/L — intrinsic property, length/area par depend nahi karti. [Q735, Q553]
 - SI unit = Ohm-meter ($\Omega \cdot \text{m}$). [Q553, Q587]
- ρ calculation: $\rho = R(\pi r^2)/L$. [Q722, Q734, Q740]
- Same R, different L aur A $\rightarrow \rho$ ratio = $(A_1/A_2) \times (L_2/L_1)$. [Q724]
- Different material, same dimension $\rightarrow \rho$ ratio = R ratio. [Q663]
- Material change: $R_{\text{new}}/R_{\text{old}} = (\rho_{\text{new}}/\rho_{\text{old}}) \times (L_{\text{new}}/L_{\text{old}}) \times (A_{\text{old}}/A_{\text{new}})$. [Q713, Q721]

24. Numerical: Series & Parallel Resistance Combinations

- Series: $R_{\text{eq}} = R_1 + R_2 + R_3 + \dots$ [Q633, Q637, Q648, Q653, Q655, Q665, Q671, Q678, Q681, Q682, Q699, Q706, Q718, Q727, Q743, Q750, Q758, Q771, Q775, Q805, Q821]
 - Same current through all resistors — R value chahe kuch bhi ho, series mein current constant rehta hai (jaise 5Ω se $6A$ guzri to 10Ω series resistor se bhi $6A$ hi guzregi). [Q637, Q648, Q665, Q671, Q727, Q771]
 - Voltage divides in ratio of resistances: $V_1/V_2 = R_1/R_2$. [Q665, Q671, Q681]
 - Cells series mein connected ho toh total EMF/potential difference = sabhi cells ki EMF ka sum hota hai (3 cells of $2V$ series mein $\rightarrow 6V$ total). [Q750]
- Parallel: $1/R_{\text{eq}} = 1/R_1 + 1/R_2 + \dots$ [Q635, Q642, Q654, Q657, Q670, Q674, Q676, Q686, Q718, Q729, Q739, Q742, Q825, Q827, Q835]
 - Same voltage across all resistors. [Q642, Q654, Q657, Q661, Q674, Q676, Q703]
 - Current divides: $I_1/I_2 = R_2/R_1$. [Q661]
 - 2 identical parallel $\rightarrow R_{\text{eq}} = R/2$. [Q642, Q674, Q676]
 - n identical parallel $\rightarrow R_{\text{eq}} = R/n$. [Q670, Q697, Q711, Q717]
- Mixed: Pehle parallel solve karo, phir series add karo. [Q635, Q686, Q707, Q827]
- Wire bent into circle $\rightarrow$ 2 semicircles in parallel $\rightarrow R_{\text{eq}} = R/4$. [Q638]
- Two resistors: $R_s = R_1 + R_2$, $R_p = R_1 R_2 / (R_1 + R_2)$. [Q733, Q737]
- Maximum resistance ke liye series; minimum ke liye parallel. [Q697, Q706, Q711, Q717, Q718, Q769, Q835]

25. Numerical: Electric Power ($P = VI = I^2R = V^2/R$)

- $P = VI$ — jab V aur I given ho. [Q652, Q660, Q674, Q675, Q676, Q702, Q730, Q757, Q777]
- $P = V^2/R$ — jab voltage aur resistance given ho (parallel mein useful). [Q643, Q692, Q694, Q731]
- $P = I^2R$ — jab current aur resistance given ho (series mein useful). [Q650, Q725, Q738]
- Same voltage (parallel) mein: $P \propto 1/R$. Lower R = higher power = brighter bulb. [Q643, Q703]
- Same current (series) mein: $P \propto R$. Higher R = higher power. [Q650]
- Resistance from power rating (same V): $R = V^2/P$. [Q649, Q709, Q720]
- Agar V double (R constant) $\rightarrow P \times 4$ ($P \propto V^2$). [Q689, Q731]
- Agar I double aur V triple $\rightarrow P \times 6$. [Q668]
- Agar dono triple $\rightarrow P \times 9$. [Q687]
- V half (R constant) $\rightarrow P = P/4$. [Q731]
- No voltage applied $\rightarrow P = 0W$. [Q741]
- $P = V/I$ galat hai — dimensionally wrong. [Q685]
- Power = Energy/Time. [Q693, Q767, Q774]
- Do bulbs parallel mein 3V source ke saath, R ratio 1:3 ho toh heat produced ratio ($H \propto 1/R$ same V par) = 3:1 (lower R wala bulb zyada heat produce karta hai). [Q645]
- 5 identical bulbs parallel mein 20V, total current 5A $\rightarrow R_{\text{total}} = V/I = 4\Omega$; example ke hisaab se har bulb ka resistance $\approx 1.25\Omega$ aur power $\approx 31.25W$ (official RRC key ke formula se). [Q729]

26. Numerical: Heat/Energy ($H = I^2Rt = VIt = V^2t/R$)

- Heat produced = Power $\times$ Time. [Q640, Q644, Q654, Q655, Q657, Q662, Q669, Q689, Q691, Q701, Q710, Q725, Q726, Q728, Q770]
- $H = I^2Rt$ — jab current direct given ho. [Q662, Q669, Q701, Q710, Q725, Q770]
- $H = VIt = V^2t/R$ — jab voltage given ho. [Q654, Q657, Q726, Q728]
- Power = Heat/Time = H/t . [Q693]

27. Numerical: Charge & Current ($Q = It$)

- Charge (Q) = Current (I) $\times$ Time (t). [Q696, Q730, Q744, Q749, Q753, Q759, Q761, Q764, Q787, Q790, Q792, Q798, Q799, Q817, Q818, Q819, Q831]
- Current (I) = Q/t . [Q794]
- Time (t) = Q/I . [Q799]
- Time hamesha seconds mein convert karo — 1 min = 60s, 5 min = 300s, 10 min = 600s. [Q753, Q759, Q761, Q764, Q787, Q790, Q792, Q794, Q798, Q817, Q818, Q819]
- Resistance value kabhi-kabhi extra information hoti hai — sirf $Q = It$ se kaam chal jaata hai. [Q764]

28. Numerical: Work Done ($W = QV$)

- Work (W) = Charge (Q) $\times$ Potential Difference (V). [Q747, Q752, Q786, Q788, Q797, Q804, Q810, Q813, Q814, Q815, Q816, Q832]
- Charge (Q) = W/V . [Q801, Q807]
- Potential Difference (V) = W/Q . [Q832 context]

- Unit hamesha Joules (J) hoti hai — N, Pa, C, A galat hain. [Q777, Q786, Q810, Q832]

29. Numerical: Energy Units (kWh ⇌ Joules)

- 1 kWh = 3.6×10^6 J (yaad rakho!). [Q755, Q762, Q766, Q785, Q811]
- 1 unit = 1 kWh = 1000 Wh. [Q644, Q754, Q768, Q773, Q776, Q782, Q785, Q800, Q802, Q803, Q806, Q811, Q824, Q834]
- Energy (kWh) = Power (kW) × Time (h). [Q640, Q644, Q754, Q765, Q768, Q776, Q782, Q800, Q802, Q803, Q824, Q834]
- Energy (Wh) = Power (W) × Time (h). [Q695, Q754, Q776, Q800]
- Joules → kWh: divide by 3.6×10^6 . [Q756, Q773, Q785, Q806, Q811]
- kWh → Joules: multiply by 3.6×10^6 . [Q756, Q766, Q773, Q785, Q806]
- Month ke liye: days × hours/day. [Q644, Q765, Q773, Q803, Q824, Q834]
- September = 30 days, October = 31, November = 30, etc. [Q644]

30. Numerical: Bulb Brightness & Power Ratings

- Parallel mein same voltage → zyada rated wattage = zyada bright. [Q643, Q703]
- Series mein same current → kam rated wattage = zyada bright (uska R zyada hota hai). [Q649]
- Current from power: $I = P/V$. [Q659, Q793, Q795, Q796]

31. Numerical: Cost Calculation & Fuse Rating

- Cost = Energy (units/kWh) × Rate per unit. [Q644: 150 kWh × Rs 6 = Rs 900]
- Fuse rating: $I = P/V$. [Q723: 2000/250 = 8A → 12A fuse inappropriate]
- Max devices from fuse: $P_{\text{total}} = V \times I_{\text{fuse}}$. [Q716: 220×15 = 3300W → 13 lamps of 100W]

32. Numerical: Induced EMF, Coulomb's Law, Magnetic Field & Electrons

- Induced EMF: $\epsilon = -N(d\Phi/dt) \rightarrow \text{EMF} \propto N$ (turns). [Q704: $N_B = 2N_A \rightarrow \epsilon$ ratio 1:2]
- n turns wali coil ka magnetic field, single loop (same radius, same current) ke comparison mein n : 1 ratio mein hota hai ($B \propto n$). [Q639]
- Straight conductor ke around magnetic field current ke directly proportional hota hai ($B \propto I$); currents 1, 2I, 3I ke liye fields $B_1 < B_2 < B_3$ (B_3 sabse zyada). [Q684]
- Induced current: $I = \text{EMF}/R$ — EMF independent of R, current depends on R. [Q688, Q708, Q712]
 - Example: $R=4\Omega$, induced current 1 mA → $\text{EMF} = IR = 1\text{mA} \times 4\Omega = 4 \text{ mV}$. [Q688]
- Coulomb's Law: $F = k(q_1q_2/r^2) \rightarrow F \propto 1/r^2$. [Q833]
 - Distance 1/5 → Force 25×. [Q833]
- Number of electrons: $n = Q/e = It/e$. [Q829, Q830]
 - $I = 1\text{A}$, $t = 1\text{s}$, $e = 1.6 \times 10^{-19} \text{ C} \rightarrow n = 6.25 \times 10^{18}$. [Q830]
- Equivalent current in hydrogen atom: $I = Qv/(2\pi R)$. [Q828]
 - $R = 5 \times 10^{-11} \text{ m}$, $v = 2.2 \times 10^6 \text{ m/s} \rightarrow I \approx 1.12 \text{ mA}$. [Q828]

33. Numerical: Wire Doubled on Itself

- Jab wire ko moodhkar aadha (doubled) karte hain:
 - Length $\rightarrow L/2$
 - Area $\rightarrow 2A$
 - New $R = \rho(L/2)/(2A) = \rho L/(4A) = R/4$. [Q784, Q808, Q809, Q822, Q823]
- Examples: $5\Omega \rightarrow 1.25\Omega$, $7\Omega \rightarrow 1.75\Omega$, $8\Omega \rightarrow 2\Omega$, $9\Omega \rightarrow 2.25\Omega$, $10\Omega \rightarrow 2.5\Omega$. [Q784, Q808, Q809, Q822, Q823]

34. Numerical: Temperature Conversion & Cross-Topic Numericals

- $K = ^\circ C + 273$. [Q826: $78^\circ C = 351K$]
- Cross-topic numerical (potential energy, mechanics ka formula jo isi Q-range mein aaya): $PE = m \times g \times h$. Example: mass=20kg, height=15m, $g=10 \text{ m/s}^2 \rightarrow PE = 20 \times 10 \times 15 = 3000 \text{ J}$ (unit Joules, na ki Watt/Newton/Pascal). [Q789]

35. ⚠ Traps / Master Formula Cheat Sheet (Numerical Q633–Q835)

- ⚠ Trap: mA ko hamesha A mein convert karo — $2.5 \text{ mA} = 2.5 \times 10^{-3} \text{ A}$. [Q646]
- ⚠ Trap: Wire stretch karne par (volume constant) $R \propto L^2$, sirf L nahi. [Q673]
- ⚠ Trap: Radius half $\rightarrow$ Area $1/4 \rightarrow R \ 4\times$. Radius $1/3 \rightarrow$ Area $1/9 \rightarrow R \ 9\times$. [Q636, Q679]
- ⚠ Trap: Series mein current same, voltage divide in R ratio. [Q665, Q671, Q681, Q771]
- ⚠ Trap: Parallel mein voltage same, current divide in inverse R ratio. [Q661]
- ⚠ Trap: Power parallel mein $P \propto 1/R$ (V same); series mein $P \propto R$ (I same). [Q643, Q650, Q703]
- ⚠ Trap: V double karne par (R constant) $P \ 4\times$ hota hai ($P \propto V^2$), $P/2$ nahi. [Q689, Q731]
- ⚠ Trap: $P = V/I$ galat hai — $P = VI$ sahi hai. [Q685]
- ⚠ Trap: 2 identical parallel $\rightarrow R_{eq} = R/2$; n identical $\rightarrow R/n$. [Q670, Q674, Q676, Q697, Q711, Q717]
- ⚠ Trap: Cost calculate karte waqt days of month yaad rakho. [Q644, Q765]
- ⚠ Trap: Power = 0 jab V = 0 (no voltage applied). [Q741]
- ⚠ Trap: V aur I dono double $\rightarrow R$ same rehta hai. [Q791]
- ⚠ Trap: Wire doubled on itself $\rightarrow R/4$ (not $R/2$). [Q784, Q808, Q809, Q822, Q823]
- ⚠ Trap: Time hamesha seconds mein — 1 min = 60s. [Q753, Q759, Q761, Q764, Q787, Q790, Q792, Q794, Q798, Q817, Q818, Q819]
- ⚠ Trap: Work done unit Joules (J) hoti hai — N, Pa, C, A galat. [Q777, Q786, Q810]
- ⚠ Trap: Resistivity (ρ) length/area par depend nahi karti. [Q735]
- ⚠ Trap: 1 kWh = $3.6 \times 10^6 \text{ J}$ — 3600 J nahi. [Q755, Q762]
- ⚠ Trap: Fuse rating hamesha thoda zyada hona chahiye actual current se. [Q723]
- ⚠ Trap: Resistance value kabhi-kabhi extra information hoti hai — $Q = It$ direct use karo. [Q764]